

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 3 SET B

SUMMER SEMESTER, 2023-2024

DURATION: 30 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. You have to design a $2^3 \times 3$ priority encoder with a Valid output. The encoder has 8 inputs labeled I_0 to I_7 , 3 output lines labeled O_2, O_1, O_0 , and a Valid signal V .

4.5
(CO5)
(PO3)

The inputs have the following randomly assigned priority (from highest to lowest):

$$\text{Priority: } I_5 > I_0 > I_3 > I_2 > I_6 > I_7 > I_1 > I_4$$

To be noted -

- If multiple inputs are **HIGH**, the encoder outputs the binary code of the **highest-priority input**.
- The **Valid** output V is 1 if any **input** is **HIGH**, and **0** if **all are LOW**. You can arrange the inputs however you like.

Fill in the truth table based on this priority.

Inputs								Outputs			
I_5	I_0	I_3	I_2	I_6	I_7	I_1	I_4	O_2	O_1	O_0	V
0	0	0	0	0	0	0	0	X	X	X	0
1	X	X	X	X	X	X	X	1	0	1	1
0	1	X	X	X	X	X	X	0	0	0	1
0	0	1	X	X	X	X	X	0	1	1	1
0	0	0	1	X	X	X	X	0	1	0	1
0	0	0	0	1	X	X	X	1	1	0	1
0	0	0	0	0	1	X	X	1	1	1	1
0	0	0	0	0	0	1	X	0	0	1	1
0	0	0	0	0	0	0	1	1	0	0	1

Solution:

Rubrics:

- 0.3 point (times 9) for each correct input row.
- 0.2 point (times 9) for each correct output row,

2. Design a combinational circuit using comparator ICs and logic gates to determine the **smallest among three input numbers**: A , B , and C . The circuit must satisfy the following specifications: 7 +
1 + 2
(CO5)
(PO3)
- **Inputs:** Three binary numbers A , B , and C (you are not required to consider the number of bits, assume they are compatible with the comparator IC).
 - **Outputs:** Three output signals:
 - **A smallest:** High (1) if $A < B$ and $A < C$
 - **B smallest:** High (1) if $B < A$ and $B < C$
 - **C smallest:** High (1) if $C < A$ and $C < B$
 - It is guaranteed that no two inputs are equal. Hence, exactly one of the outputs will be high at a time.
 - **Comparator IC:** You may only use 2-input comparator blocks. Each comparator has two n -bit number inputs (X and Y) and three outputs: $X > Y$, $X < Y$, and $X = Y$, where X and Y are two n -bit numbers.
You are not required to design the comparator IC, it may be represented as a block in your diagram.
 - **Design Requirement:** Using these comparator IC blocks and any other component as you wish, draw the **block diagram** of a combinational circuit that produces the required outputs. You may assume ideal behavior and unlimited availability of logic gates and comparators.

Note: A block diagram means that you do not need to draw the internal structure of the comparator ICs. Instead, use labeled blocks showing inputs and outputs clearly.

Answer the following questions -

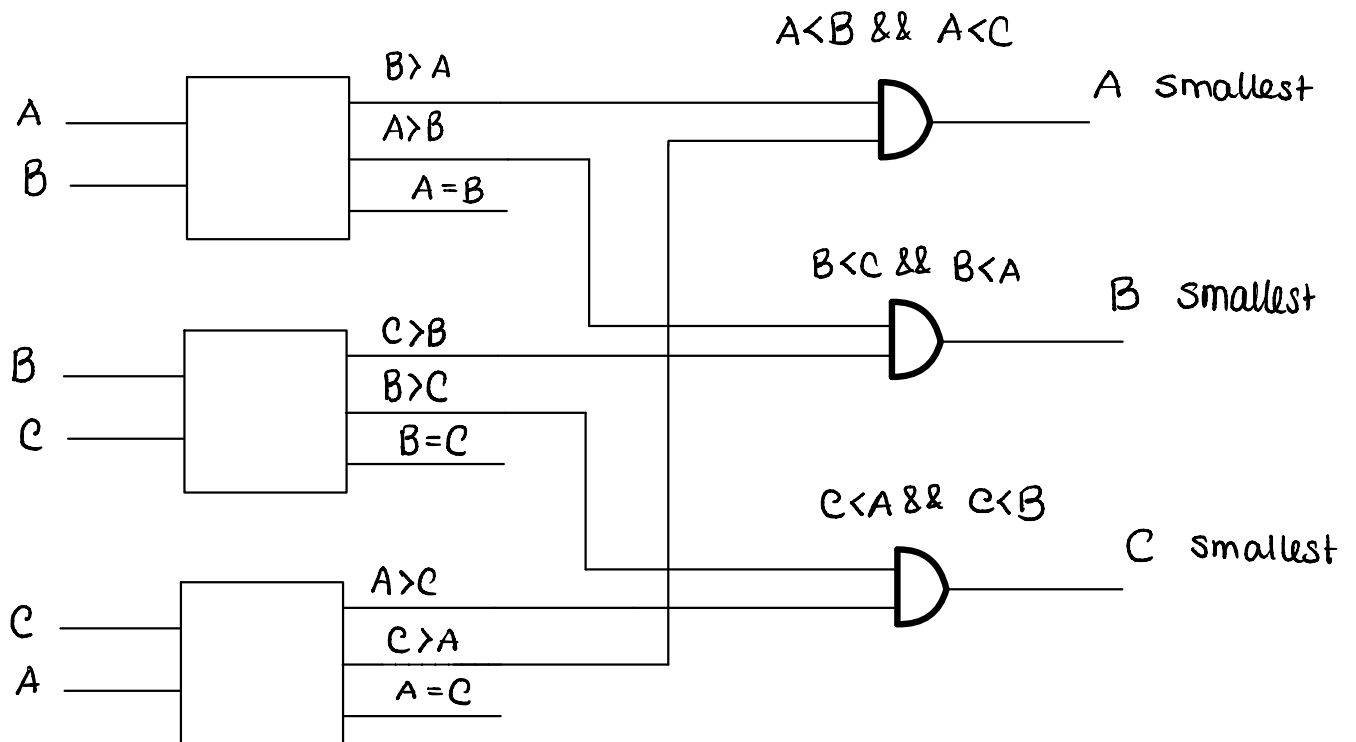
- a) Design the above-mentioned circuit.
- b) Explicitly mention the number of 2-number comparator ICs you have used.
- c) Briefly explain the working mechanism (avoid unnecessary details).
- d) Propose the most efficient solution in terms of the number of comparators used. **[Bonus 5 Marks]**

Solution:

Rubrics:

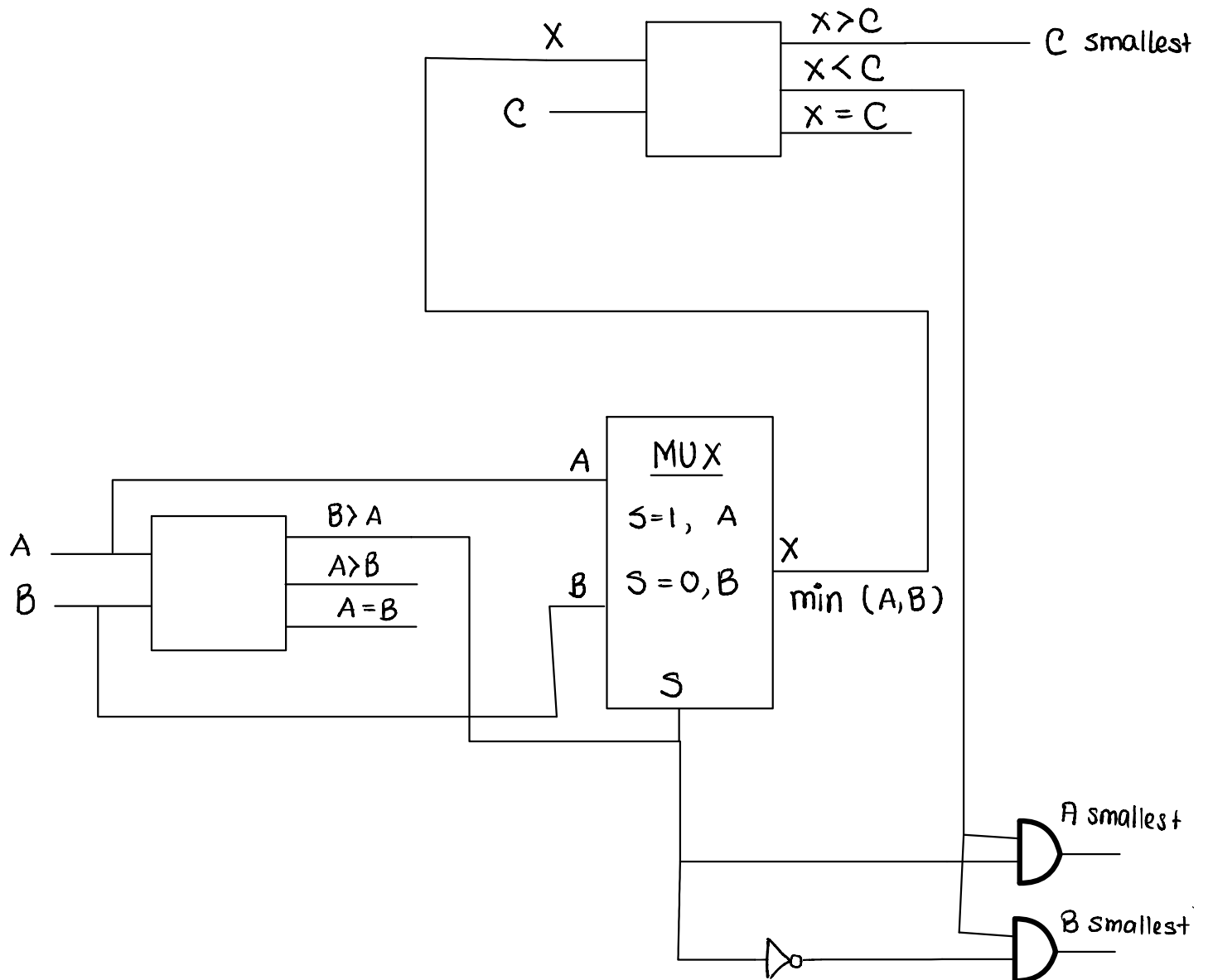
- 2 points (times 3) for comparison between each pair of input.
- 1 points for correctly determining the output for each output.
- 1 points for correctly mentioning the correct number of comparators used.
- 2 points for accurate logical justification.
- **5 bonus points** for efficient solution using only 2 comparators.

Naïve Approach



Comparators used: 3

Efficient Solution



Comparators used : 2

3. Given,

3.5 + 2

$$F_1(A, B, C) = \sum m(3, 5, 7)$$

(CO5)
(PO3)

$$F_2(A, B, C) = \sum m(2, 3, 6)$$

Complete the PLA circuit for F_1 and F_2 by making the necessary connections (using 'X' at junctions) using the following **PLA (Programmable Logic Array)** structure.

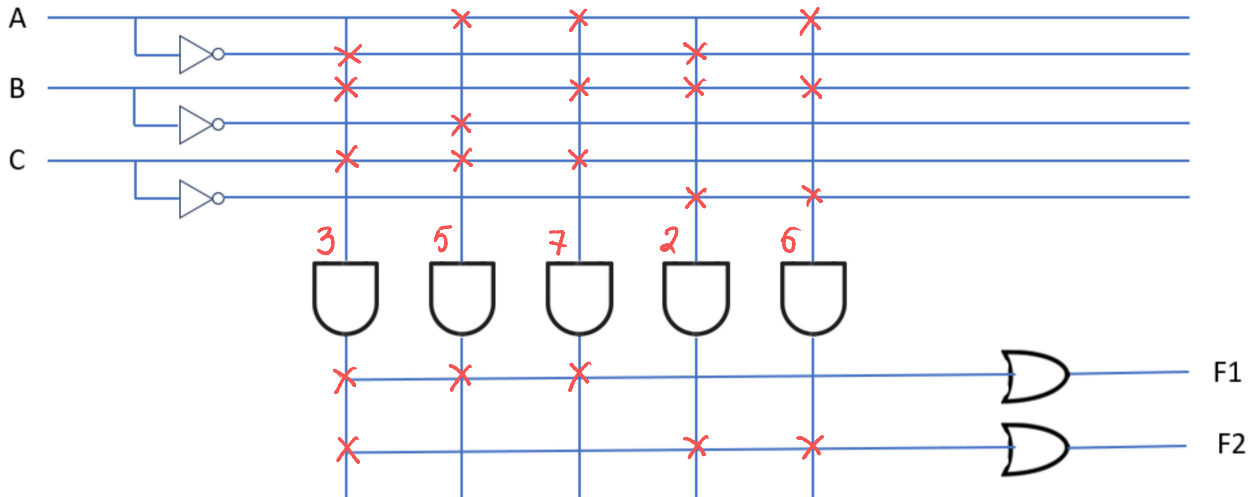


Figure 1: PLA Circuit

Solution:

Rubrics:

- 0.5 point (times 5) for each correct AND array combination.
- 0.75 point (times 2) for each correct OR array output.
- 0.5 point each (times 3) for correctly defining n , m and p and stating their values.
- 0.25 point (times 2) for each correct MCQ answer.

And answer the following questions-

a) A PLA circuit structure is defined by $n \times p \times m$. Write down what each symbol represents and provide their values according to the given PLA circuit diagram that you have constructed.

- n : Input, Value: 3
- p : Intermediate Output of AND array, Value: 5
- m : Output of OR array, Value: 2

b) Put a tick next to the correct option.

For a **ROM (Read Only Memory)**:

- AND Array: ☒ Fixed ☐ Programmable
- OR Array: ☐ Fixed ☒ Programmable